Product Requirements Document (PRD)

PersonaReview AI: Personalized Review Summarization Engine

Document Version: 1.0
Status: Ready for Review
Target Delivery: Q3-Q4
Primary Stakeholders: Product, AI/ML Engineering, UX/UI, Data Platform, Privacy & Trust & Safety

1. Executive Summary & Problem Statement

1.1 Executive Summary

Current e-commerce and product review platforms provide either unorganized, high-volume customer reviews or one-size-fits-all AI-generated summary bullet points (e.g., "Customers like the sound quality, but some complain about battery life").

PersonaReview AI dynamically synthesizes, filters, and summarizes product reviews tailored directly to each prospective buyer's distinct preferences, context, and purchasing priorities (e.g., durability, ease of setup, price-to-performance, compatibility, or specific use cases).

1.2 Problem Statement

- **Information Overload:** Popular products contain thousands of reviews. Reading through them causes severe cognitive fatigue and decision paralysis.
- **Irrelevance of Generic Summaries:** A parent buying headphones for a toddler cares about volume limiters and durability; an audiophile cares about DAC support and frequency response. Generic summaries dilute the signal for both.
- **Lost Conversion & High Return Rates:** When shoppers cannot verify whether a product meets their specific niche needs, they either abandon the cart or purchase with mismatched expectations, driving costly returns.

2. Target Personas & User Journeys

2.1 Target Personas

Persona	Description	Key Motivations & Needs
P1: The Attribute-Specific Buyer (Explicit)	Shoppers with strict criteria (e.g., apartment size, sensitive skin, Linux compatibility).	Wants to see immediate evidence regarding their specific constraints without searching through 2,000+ reviews.
P2: The Habitual Returner / Value Seeker	Shoppers who prioritize build longevity, reliability, and true fit over cosmetic hype.	Wants verified long-term ownership feedback (3+ months) filtered to quality and durability.
P3: The Casual / Cold-Start Browser	New visitors with no prior purchase history on the platform.	Needs clear, interactive preference selector chips to guide review synthesis on the fly.

2.2 End-to-End User Journey

3. Goals, Objectives & Key Results (OKRs)

3.1 Business Goals

- **Increase Add-to-Cart (ATC) & Conversion Rate:** Reduce time-to-decision by surfacing relevant sentiment instantly.
- **Decrease Return Rates:** Better align customer expectations with product reality before checkout.
- **Boost Engagement:** Increase interaction depth on Product Detail Pages (PDP).

3.2 Success Metrics (KPIs)

Metric	Target	Measurement Method
Conversion Rate (CVR) Uplift	$\geq +3.5\%$≥ +3.5%	A/B test against standard generic review summary
Time-to-Purchase (PDP to Checkout)	-20%−20% reduction	Session analytics on PDP interaction duration
Post-Purchase Return Rate (Mismatch reasons)	-5.0%−5.0% decrease	Return reason tracking ("not as expected")
Summary Faithfulness / Grounding	$\geq 98\%$≥ 98% hallucination-free	Automated RAG triad evaluation (G-Eval / TruLens) + Human spot audits
p95 Latency (Cached & Streaming)	$< 350\ \text{ms}$< 350 ms TTFT	Edge / API gateway latency metrics

4. Customer Preference Profiling Engine

The personalization engine operates along two primary dimensions: **Implicit Preferences** and **Explicit Preferences**, balanced with strict privacy and freshness decay.

4.1 Preference Signals

1. **Explicit Inputs (Priority 1):**
    - Active on-page attribute chips (e.g., "Focus on: Battery Life, Noise Cancellation").
    - User account preference settings (e.g., size fit profile, allergies/dietary tags, tech proficiency).
2. **Implicit Signals (Priority 2):**
    - Past categories purchased and returned with aspect-level return feedback.
    - Recent search keywords (e.g., searched for "heavy bass waterproof speaker").
    - Dwell time on specific review categories.
3. **Cold-Start Strategy:**
    - Display the top 5 most frequently discussed product attributes as clickable pills directly above the summary card.

- Automatically fall back to the top 2 overall positive and 1 critical attribute until the user interacts.

5. System Architecture & AI Pipeline

5.1 Pipeline Steps

1. **Aspect & Sentiment Tagging (ABSA):**
   - Every review is decomposed into tuples: `(Aspect, Sentiment, Confidence, Quote, Time, VerifiedPurchase)`.
   - Examples of aspects: `Battery Life`, `Build Quality`, `Setup Ease`, `Customer Service`.
2. **Preference-Weighted Retrieval (RAG):**
   - Given user preference vector UU and product PP, retrieve review chunks matching top NN user-weighted aspects.
   - Negative sentiment chunks for user-prioritized attributes are **always included** to ensure objective balance and trust.
3. **Constrained Generation:**
   - The LLM synthesizes structured bullet points adhering to a strict schema:
     - **Headline:** 1-sentence bottom line tailored to the user's primary criteria.
     - **What matches your priorities:** Key positive takeaways with inline citation numbers `[1]`, `[2]`.
     - **What to watch out for:** Critical warnings relevant to the user's focus areas.
4. **Attribution & Grounding Verification:**
   - Any assertion made by the model must link to an existing verified review ID. Sentences without grounding are rejected by the validator.

6. Functional Requirements

ID	Feature	Priority	Description	Acceptance Criteria
FR-01	**Personalized Summary Card**	**P0**	Dynamic UI card placed directly above the review section on PDP.	Displays: 1 summary sentence, 2-3 matched aspect pros, 1-2 balanced caveats, and clickable citations.
FR-02	**Interactive Aspect Customizer**	**P0**	Clickable pills/tags allowing user to change or add priorities on the fly.	Clicking an aspect re-ranks or re-synthesizes the summary in $< 400\ \text{ms}$< 400 ms (utilizing pre-computed cluster cache).
FR-03	**Citation & Source Transparency**	**P0**	Inline reference tags `[1]`, `[2]` linking to actual review quotes.	Clicking opens a drawer or scrolls smoothly to highlighted original review text with verified purchaser badge.
FR-04	**"Why am I seeing this?" Explainer**	**P1**	Micro-copy and tooltip disclosing personalization rationale.	Tooltip states: "*Tailored based on your interest in [Attribute X] and your recent searches.*" Includes reset button.
FR-05	**Review Freshness & Recency Weighting**	**P1**	Product revisions/firmware update awareness.	Flags reviews posted before major product updates; weights recent reviews (last 6-12 months) higher.
FR-06	**Explicit Opt-out / Privacy Toggle**	**P1**	User setting to disable personalization.	Toggling off displays standard aggregate summary; clears session preference cache immediately.
FR-07	**Sentiment Discrepancy Alert**	**P2**	Proactive warning if high-priority aspect has polarized feedback.	Shows warning indicator: "*Notice: Shoppers with your preference had mixed experiences regarding [Durability].*"

7. Non-Functional & Technical Requirements

7.1 Latency & Performance Budget

- **Cached Personalization (Common Personas):** $< 150\ \text{ms}$< 150 ms p95.
- **Dynamic Retrieval + LLM Streaming (TTFT):** $< 450\ \text{ms}$< 450 ms p95; full response within $1.2\ \text{s}$ 1.2 s.

- **Caching Architecture:** Pre-cluster reviews by the top 10 most common aspect combinations per product. Serve pre-compiled summaries for known clusters, avoiding real-time LLM calls for 80% of traffic.

7.2 Scalability & Cost Optimization

- Tiered LLM approach:
    - Fast, smaller distilled model (e.g., fine-tuned 8B parameter model) for Aspect Tagging & Extraction.
    - Efficient reasoning model (or quantized low-latency endpoint) for real-time customer-conditioned synthesis.
- Rate limits: Max 5 dynamic re-summaries per session per user to prevent API abuse.

7.3 Data Privacy, Security & Compliance

- **Compliance:** Full compliance with GDPR, CCPA, and CPRA.
- **Data Minimization:** No personal health information (PHI) or sensitive PII passed into LLM prompt contexts.
- **Zero Data Retention Agreement:** Enterprise LLM endpoints must enforce zero data retention (no model training on customer browsing or preference data).

8. Trust, Safety & Anti-Hallucination Guardrails

> IMPORTANT
>
> Personalization must **never** create a misleading "sales pitch" by filtering out genuine negative reviews. Credibility and trust are paramount.

1. **Mandatory Critical Sentiment Quota:**
    - If $\geq 15\%$≥ 15% of reviews mention a defect or negative experience in an aspect the user prioritizes, the summary **must** include a "Caveats to Consider" section.
2. **Review Authenticity Filtering:**
    - Filter out incentivized, unverified, or bot-flagged reviews using the platform's review spam score before vector indexing.
3. **Hallucination Blocker:**

- Fact-checking layer checks exact numerical claims (e.g., "Lasts 12 hours on battery") against product specs and review citations. If inconsistent, the claim is excised.

9. UX / UI Specification & Wireframe Blueprint

9.1 Summary Component Layout (PDP)

```
+-------------------------------------------------------------------------------+
|  PersonaReview AI   [ Personalized for: Battery Life | Noise Isolation ] (Edit)|
+-------------------------------------------------------------------------------+
|  "Ideal for daily commuters who need reliable battery and commute-ready ANC,  |
|   though some note the ear cups feel warm during extended workouts."          |
|                                                                               |
|  Matches Your Priorities:                                                     |
|  * Battery Life: Average user reports 28-30 hrs on single charge [1][3].      |
|  * Noise Cancellation: Blocks subway noise effectively, though voice leaks [2]|
|                                                                               |
|  Things to Keep in Mind:                                                      |
|  ! Comfort: Users with larger ears report clamping tightness after 2+ hrs [4] |
|                                                                               |
|  [Show 14 reviews matching your focus]     [View Standard Summary] [Why this?]|
+-------------------------------------------------------------------------------+
```

10. Phased Roadmap & Milestones

Phase 1: MVP (8 Weeks)

- Implement Aspect-Based Sentiment Analysis (ABSA) across top 5 categories.
- Deliver interactive on-page UI chips (Explicit preference only, cold-start).
- Rule-based aspect filtering with standard LLM synthesis + strict citation grounding.
- Internal dogfooding and offline faithfulness evaluation.

Phase 2: Personalization & Implicit Signals (6 Weeks)

- Integrate user browsing/purchase history vector profiling (Implicit preference).
- Implement cluster-level summary caching to achieve $< 250\ \text{ms}$ < 250 ms latency at scale.
- A/B testing launch on 20% PDP traffic against generic summary control.

Phase 3: Advanced Intelligence & Expansion (8 Weeks)

- Real-time product update and revision awareness (version-specific feedback).
- Multimodal review integration (surfacing user photos/videos relevant to user's prioritized attribute).
- Platform-wide rollout across all catalog categories.

11. Open Questions & Key Decisions Needed

1. **Implicit Signal Threshold:** How strongly should past browsing weigh against current category norms? (Recommended default: 70% explicit on-page signals, 30% implicit historical signals).
2. **Product Revision Boundaries:** When a manufacturer releases a hardware revision (V2), how quickly should old reviews be phased out of the personalized summary context?
3. **Opt-in vs. Opt-out Default:** Should personalized review synthesis be opt-in (click to personalize) or default-on with an opt-out toggle?